

Solve the following equations:

$$\ln(x+1) - \ln(-2x+3) = 2\ln 2 + \ln 3$$

Solution:

$$\ln\left(\frac{x+1}{-2x+3}\right) = \frac{\ln(12)}{1}$$

$$\text{then: } e^{\ln\left(\frac{x+1}{-2x+3}\right)} = e^{\ln(12)}$$

$$\frac{x+1}{-2x+3} = 12$$

$$x+1 = (-2x+3)(12)$$

$$x+1 = -24x+36$$

$$25x = 35$$

$$x = \frac{35}{25} = 1.4 \Rightarrow$$

$$\left\{ \begin{array}{l} x+1 > 0 \\ x > -1 \end{array} \right\} \left\{ \begin{array}{l} -2x+3 > 0 \\ -2x > -3 \\ x < \frac{3}{2} \end{array} \right\}$$
$$-1 < x < \frac{3}{2}$$

$$\boxed{x = \frac{7}{5}}$$

$$e^{2x} + 4e^x - 5 = 0$$

$$\text{Let } e^x = t$$

$$\Rightarrow t^2 + 4t - 5 = 0$$

$$t_1 = 1 \text{ and } t_2 = -5$$

$$\Rightarrow e^x = 1$$

OR $e^x = -5$ rejected

$$\text{then: } e^0 = 1$$

$$x = 0$$

$$e > 0$$

$$\text{So: } x = 0$$

